

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR**Odd Semester Mid-term Examination, 2025-26**

Course Code: CEC302

Full Marks: 25

Course Name: Construction, Materials, & Concrete Technology

Time: 90 Mins

Question Paper No.: NITDGP/CEC302

Date of Exam: 25/09/2025

Instructions: Answer all the questions.

Materials to be supplied: Graph paper shall be supplied, if required.

PART - A

Question No.	Body of the Question	Marks	Mapped CO
1	What do you understand by the following principles of planning a building: (a) Circulation, (b) Flexibility?	2	CO2
2	How the following factors affect the orientation of a building: (a) Direction of road / street, (b) Wind direction.	2	CO2
3	Differentiate between the following with a neat sketch: Set-back line & Building line.	2	CO3
4	What do you understand by (a) Head room of a room and (b) Head room of a stair? Show with sketches.	2	CO2
5	What are the different types of settlement of soils? How are they associated with sandy & clayey soils?	2	CO3

PART - B

PART - B																			
Question No.	Body of the Question	Marks	Mapped CO																
16	a) What do you understand by "initial and final setting time" of cement? b) Explain the procedure to determine the initial and final setting time of cement.	5	CO1																
17	Explain the term "fineness modulus" of aggregate with its significance. Determine the fineness modulus of the aggregate from the information provided below. <table><thead><tr><th>Sieve Size (mm)</th><th>Weight retained on it (g)</th></tr></thead><tbody><tr><td>4.75</td><td>5.7</td></tr><tr><td>2.36</td><td>3.8</td></tr><tr><td>1.18</td><td>8.6</td></tr><tr><td>0.600</td><td>42.2</td></tr><tr><td>0.300</td><td>193.1</td></tr><tr><td>0.150</td><td>35.6</td></tr><tr><td>Pan</td><td>11.0</td></tr></tbody></table>	Sieve Size (mm)	Weight retained on it (g)	4.75	5.7	2.36	3.8	1.18	8.6	0.600	42.2	0.300	193.1	0.150	35.6	Pan	11.0	5	CO1
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Pan	11.0																		
18	Explain the functions of the following ingredients in brick earth on bricks. a) Silica b) Alumina c) Oxides of iron	5	CO1																

Course Outcomes

CO1: Apply the knowledge about building materials for construction practices.
 CO2: Understand the building components and planning
 CO3: Learn the basic principles of construction

NITDGP/BTECH/Reg/Odd/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Odd Semester End-term Examination, 2025-26

Course Code: CEC302

Course Name: Construction, Materials, & Concrete Technology

Question Paper No.: NITDGP/CEC302

Full Marks: 60

Time: 180 Mins

Date of Exam: 25/11/2025

Instructions: Answer all the questions.

Materials to be supplied: Graph paper shall be supplied, if required.

PART - A

Question No.	Body of the Question	Marks	Mapped CO
1	(a) Describe the features of the following with neat sketches: (i) Raft foundation (ii) Well foundation (iii) Compare the two types of R.C.C. piles.	(3+3)+4 = 10	CO3
2	(a) Show the uses of the following types of walls with neat sketches of their typical sectional elevations: (i) Parapet wall, (ii) Sleeper wall, (iii) Buttressed wall (iv) Partition wall. (b) What are the points to be considered for supervising construction of brick masonry?	(2x3) + (2x2) = 10	CO2
3	(a) Discuss the various methods of damp proofing with their places of application. (b) What do you understand by the following? Show with neat sketches. (i) Line of nosing, (ii) Newel, (iii) Balustrade, (iv) B.F.S, (v) Weathered pointing.	5+5 = 10	CO3

PART - B

Question No.	Body of the Question	Marks	Mapped CO
4	Write a short note on the followings: (a) Bogue's Compound (b) Ettringite	2.5 x 2 = 5	CO1
5	(a) Classify the aggregates on the basis of unit weight. (b) What do you understand by flakiness and elongation index of aggregate?	2 + 3 = 5	CO1
6	(a) Explain the following terms: (i) Squint bricks (ii) Coping bricks) (b) Dimension of both the faces of one brick is observed as 190 mm x 90 mm x 90 mm. During the crushing strength test, the	2 + 3 = 5	CO1

Course Outcomes

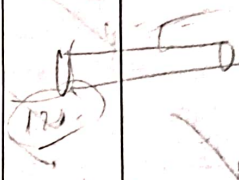
CO1: Apply the knowledge about building materials for construction practices.

CO2: Understand the building components and planning

CO3: Learn the basic principles of construction

16
T00x500

Ans = 0.7406

	brick fails when 225 kN load is applied. Determine the crushing strength of the brick and comment on the quality of the brick.		
7	(a) Explain the term "Workability of concrete". (b) Name the experiments to measure the workability of concrete and explain any one in details. <i>compaction factor</i>	2 + 3 = 5	CO1
8	(a) With the help of neat sketches, explain the relationship between Compressive strength of concrete and Water-cement ratio. (b) Concrete specimens having dimension 100 mm × 100 mm × 500 mm were tested to determine flexural strength of concrete. Average maximum load and distance between fracture line to the nearer support were observed as 16 kN and 12.4 cm respectively. Determine the flexural strength of concrete.	3 + 2 = 5 	CO1
9	(a) Differentiate between "Nominal mix" and "Design mix". (b) Enlist the data/information required for mix proportioning of a particular grade of concrete as Indian standard practices.	2 + 3 = 5	CO1